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1 Geothermal Energy in Rwanda: Summary of Findings

1.1 Overview

Rwanda has significant geothermal energy potential as part of its strategy to diversify energy sources, reduce fossil fuel dependence, and achieve universal electricity access.

1.2 Key Potential Estimates

- **Conservative estimate**: Up to 100 MW (Rwanda Energy Group, African Geothermal Association)
- **Higher estimates**: Some sources suggest potential exceeding 700 MW (Rwanda Dispatch)

1.3 Prospect Zones

1.3.1 Northwestern Volcanic Area

- Gisenyi

- Karisimbi
- Kinigi

1.3.2 Southern Rift-Fault Area

- Bugarama

1.3.3 Additional Areas Mentioned

- Near Lake Kivu
- Volcanoes National Park
- Rubavu Graben
- Mashyuza

1.4 Exploration History

- **1980s**: Early geothermal work began
- **2006**: Two deep wells drilled at Karisimbi (no viable system found)
- **2006-2012**: Early surface studies identified four prospects
- **Current focus**: Detailed surface investigations in Bugarama and Gisenyi, with gradient-well work in Bugarama

1.5 Development Plans (Per 2015 Master Plan)

- **5-year exploration phase**: Surface work, drilling, well testing
- **3-year construction phase**: If resources proven
- Rwanda planned to host the Seventh Africa Rift Geothermal Conference (ARGeo-C7) in October 2018

1.6 Strategic Importance in National Energy Policy

1.6.1 Electricity Access Goals

- Target: 100% electricity access by 2024
- Breakdown: 52% on-grid, 48% off-grid

1.6.2 Capacity Expansion Targets

- Increase installed capacity from 216 MW to 512 MW by 2024
- Geothermal positioned as clean, reliable baseload power

1.6.3 Climate Commitments

- Critical for meeting Rwanda's 2030 emissions-reduction pledge of 38%
- Geothermal provides "stable and non-emitting power" compared to other renewables

1.7 Investment Requirements

- **$600 million**: Estimated needed for universal electricity access
- **$11 billion**: Over next decade for mitigation and adaptation
- Private sector and financial institutions viewed as essential partners for development

1.8 Regional Cooperation and Lessons from Kenya

Rwanda can significantly accelerate its geothermal development by leveraging Kenya's extensive experience, as Kenya has emerged as a global leader with over 900 MW of installed geothermal capacity (primarily from the Olkaria complex) as of 2026.

1.9 Key Lessons from Kenya's Geothermal Success:

1. **Phased Exploration Approach**: Kenya uses slim holes (6-8.5") for initial exploration, followed by production wells (9.5-12.25") after resource confirmation, reducing wasted expenditure.
2. **Directional Drilling**: Targeting specific fracture zones increases well productivity by 30-50% compared to vertical wells.
3. **Reservoir Management**: Sophisticated monitoring including pressure/temperature tracking, tracer testing, and production profiling optimizes steam field longevity.
4. **Power Plant Standards**: Modular designs (150-350 MW units) allow for phased expansion and easier maintenance.
5. **Financial Mechanisms**: Partial risk guarantees from development partners (World Bank, AfDB) cover exploration drilling risks, while inflation-indexed PPAs provide revenue certainty.

1.10 Recommended Kenya-Rwanda Cooperation Framework:

- **Technical Exchange Program**: Formal partnerships with KenGen and GDC for knowledge transfer
- **Training Initiatives**: Develop local Rwandan capacity through Kenyan training programs
- **Joint Ventures**: Explore partnership models for initial drilling and plant development
- **Regional Drilling Hub**: Leverage Kenyan drilling contractors to establish East African expertise
- **Financing Collaboration**: Joint applications to climate finance facilities (Green Climate Fund, etc.)

By implementing these lessons and fostering regional cooperation, Rwanda can reduce technical risks, improve project economics, and accelerate the timeline for commercial geothermal production.

1.11 Current Status (as of research completion)

- Exploration ongoing with focus on surface investigations
- No commercial geothermal production yet despite significant potential
- Test sites planned at Kinigi and Karisimbi for future development

1.12 Sources Consulted

1. Rwanda Energy Group - Geothermal Section
2. African Geothermal Association - Rwanda Profile
3. UNIDO Geothermal Fact Sheet for Rwanda (PDF)
4. Rwanda Dispatch - Geothermal Schemes Article

Note: This summary compiles information from multiple sources researched on September 6, 2026.

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2 Hot Springs Energy Profitability Analysis for Rwanda

2.1 Executive Summary

This analysis evaluates the profitability and return on investment (ROI) potential of harnessing Rwanda's hot springs energy for both electrical generation and direct thermal applications (particularly water pumping). Based on available data from Rwanda Energy Group, JICA reports, and engineering assessments, hot springs energy presents viable opportunities with varying timelines for ROI depending on application type.

2.2 Rwanda's Hot Springs Resources

2.2.1 Temperature and Location Data

- **Gisenyi Hot Springs**: 70-75°C (confirmed from multiple sources)
- **Karisimbi, Kinigi, Bugarama**: Documented as geothermal prospects (specific temperatures not publicly available in accessible sources)
- **Primary Prospect Zones**:
  - Northwestern Volcanic Area: Gisenyi, Karisimbi, Kinigi
  - Southern Rift-Fault Area: Bugarama

2.2.2 Resource Potential

- **Electrical Generation Potential**: Up to 100 MW (per Rwanda Energy Group and 2015 Master Plan)
- **Temperature Range for Direct Use**: 70-75°C+ (suitable for various direct applications)

2.3 Energy Demand Context: Rwanda's Current Needs

2.3.1 Electricity Demand

- **Current Installed Capacity**: ~216 MW (from MININFRA)
- **Target Capacity by 2024**: 512 MW (growth of ~296 MW needed)
- **Electricity Access Goal**: 100% by 2024 (52% on-grid, 48% off-grid)
- **Current Access**: 51% of households (37% grid, 14% off-grid)

2.3.2 Water Pumping Energy Requirements

From WATER_PUMPING_ENERGY analysis: - **Energy Required**: 0.64-0.89 kWh/m³ (depending on efficiency and friction losses) - **Kigali's Estimated Daily Water Demand**: ~100,000 m³/day - **Annual Electricity for Pumping**: ~32.5 GWh/year - **Annual Pumping Cost**: ~$5.2 million USD/year (at Rwanda's tariff of ~$0.16/kWh)

2.4 Profitability Analysis by Application Type

2.4.1 1. Electrical Power Generation

2.4.1.1 Technical Feasibility

- **Resource Temperature**: 70-75°C+ (suitable for binary cycle power plants)
- **Technology Appropriateness**: Binary cycle ORC (Organic Rankine Cycle) plants ideal for <150°C resources
- **Plant Efficiency**: ~10-15% for low-temperature geothermal (vs. 10-20% for high-temperature flash plants)

2.4.1.2 Economic Assessment **Capital Costs (Estimated)**: - Binary Cycle Geothermal Plant: $2,500-$4,000 per kW installed - 100kW Plant: $250,000-$400,000 - 1MW Plant: $2.5-$4.0 million - 5MW Plant: $12.5-$20.0 million - 10MW Plant: $25.0-$40.0 million

Operating Costs: - O&M: 1-3% of CAPEX annually - Minimal fuel costs (geothermal heat is free) - Parasitic load: ~10-15% of gross generation

Revenue Streams: - Electricity Sales: At Rwanda's tariff (~$0.16/kWh for commercial/industrial) - Potential for PPAs (Power Purchase Agreements) with utilities or industries - Carbon credits: Potential additional revenue from emissions reduction

ROI Calculations for Electrical Generation: *Assumptions: 85% capacity factor, 25-year lifespan, 0.5% annual degradation*

Plant Size	CAPEX Estimate	Annual Generation	Annual Revenue	Simple Payback	25-Year NPV (8% discount)
100kW	$325,000	744,600 kWh	$119,136	2.7 years	$1.1M-$1.5M
500kW	$1.6M	3.7M kWh	$595,680	2.7 years	$5.5M-$7.5M
1MW	$3.25M	7.4M kWh	$1.19M	2.7 years	$11M-$15M
5MW	$16.25M	37.2M kWh	$5.95M	2.7 years	$55M-$75M
10MW	$32.5M	74.4M kWh	$11.9M	2.7 years	$110M-$150M

Note: These estimates assume binary cycle technology and exclude exploration/drilling risk costs.

2.4.2 2. Direct Thermal Applications (More Immediate ROI Potential)

Direct use of geothermal heat (without electricity conversion) offers significantly better ROI due to: - Higher efficiency (70-90% vs 10-15% for electricity generation) - Lower capital costs (no power conversion equipment needed) - Immediate displacement of expensive thermal fuels

2.4.2.1 Applicable Direct Use Scenarios for Rwanda: A. Water Pumping Assistance (Most Promising) - Concept: Use geothermal heat to reduce electrical energy needed for water pumping - **Methods**: 1. **Thermal Boosting**: Pre-heat water to reduce pumping work (limited applicability) 2. **Absorption Cooling/Heating**: For HVAC in water treatment facilities 3. **Hybrid Systems**: Geothermal-assisted heat pumps for pumping stations 4. **Direct Steam Use**: If sufficient pressure/temperature available (unlikely at 70-75°C)

B. Greenhouse Heating for Agriculture - **Temperature Suitability**: 70-75°C ideal for greenhouse heating - **Applications**: - Vegetable production (tomatoes, peppers, cucumbers) - Flower cultivation (roses, carnations for export) - Seedling nurseries - **Economic Benefits**: - Displace diesel/biomass heating ($0.50-$1.00/liter diesel equivalent) - Extended growing seasons - Higher yields and quality - Potential export market access

C. Aquaculture (Fish Farming) - **Temperature Suitability**: 70-75°C suitable for tropical fish species - **Applications**: - Tilapia farming (optimal 28-30°C, requires mixing) - Catfish cultivation - Prawn/shrimp farming - **Benefits**: - Faster growth rates - Year-round production - Reduced disease incidence - High-value protein production

D. Industrial Process Heat - **Applications**: - Food processing (pasteurization, drying, sterilization) - Textile industry (dyeing, drying) - Breweries (mashing, wort boiling) - Dairy processing (pasteurization) - **Temperature Match**: 70-75°C suitable for many low-to-medium temperature processes

2.4.2.2 Direct Use ROI Analysis

Assumptions: - Direct thermal efficiency: 75% (heat exchanger + distribution losses) - Displaced fuel: Diesel at 1,700 RWF/L (~$1.50/L) with 10 kWh/L thermal equivalent - Effective cost of displaced heat: ~$0.15/kWh thermal - Geothermal heat cost: Near zero (only pumping/capital costs)

Capital Cost Estimates: - Direct Use System (heat exchangers, piping, controls): $500-$1,500 per kW thermal - 100kW thermal system: $50,000-$150,000 - 500kW thermal system: $250,000-$750,000 - 1MW thermal system: $0.5-$1.5 million

Annual Savings Calculation: *For a 500kW thermal system operating 8,000 hours/year:* - Thermal energy delivered: 500 kW × 8,000 h × 0.75 efficiency = 3,000,000 kWh thermal/year - Displaced diesel: 3,000,000 kWh ÷ 10 kWh/L = 300,000 L diesel/year - Cost savings: 300,000 L × 1,700 RWF/L = 510,000,000 RWF/year (~$443,000 USD/year) - **Simple Payback**: 0.5-1.5 years (based on CAPEX range)

2.4.3 Comparison: Electrical vs. Direct Use ROI

Metric	Electrical Generation	Direct Thermal Use
Typical Efficiency	10-15%	70-90%
CAPEX per Useful kW	$2,500-$4,000/kWe	$500-$1,500/kWt
Simple Payback	2.5-4.0 years	0.5-2.0 years
Technical Complexity	High (power plant, grid connection)	Low-Medium (heat exchangers, pumps)
O&M Requirements	Moderate-Specialized	Low-Moderate
Scalability	Modular but significant infrastructure	Highly scalable, distributed applications
Risk Profile	Higher (exploration, drilling, power conversion)	Lower (primarily surface equipment)

2.5 Highest Potential for Success and Fastest ROI

2.5.1 Ranked Opportunities for Rwanda:

1. **Greenhouse Agriculture Heating (Fastest ROI)**
    - **Why**: Direct thermal match, high-value outputs, established markets

- **ROI Timeline**: 6-18 months
- **Scalability**: Modular, can start small (100kW thermal) and expand
- **Additional Benefits**: Food security, export earnings, job creation
- **Temperature Match**: Excellent (70-75°C ideal for many crops)

2. **Aquaculture Facility Heating**
   - **Why**: Strong domestic and export market for fish, temperature suitability
   - **ROI Timeline**: 12-24 months
   - **Scalability**: Can integrate with existing fish farms
   - **Additional Benefits**: Protein production, diversification of agriculture
3. **Industrial Process Heat Applications**
   - **Why**: Displace expensive diesel/heavy fuel oil in industries
   - **ROI Timeline**: 18-36 months
   - **Scalability**: Depends on industrial park development
   - **Target Industries**: Food processing, textiles, dairy, breweries
4. **Geothermal-Assisted Heat Pumps for Water Facilities**
   - **Why**: Addresses significant municipal energy cost (water pumping ~$5.2M/year for Kigali)
   - **ROI Timeline**: 2-4 years
   - **Scalability**: Can be implemented at pumping stations and treatment plants
   - **Technology**: Use geothermal as heat source/sink for high-efficiency heat pumps
5. **Binary Cycle Electrical Generation**
   - **Why**: Contributes to grid stability and renewable energy targets
   - **ROI Timeline**: 3-5 years (after exploration risk is mitigated)
   - **Scalability**: Limited by resource size and transmission access
   - **Best For**: Larger prospects (>1MW) near grid infrastructure

2.6 Implementation Recommendations for Shyuha Project

2.6.1 Phase 1: Direct Use Pilots (0-18 months)

- **Target**: Greenhouse heating or aquaculture at Gisenyi or Bugarama sites
- **Actions**:
  1. Conduct detailed temperature and flow rate measurements at hot springs
  2. Engage with agricultural cooperatives or fish farmers
  3. Design and install pilot direct use system (50-200kW thermal)
  4. Monitor performance and ROI
  5. Develop business model for scale-up

2.6.2 Phase 2: Industrial and Municipal Applications (12-36 months)

- **Target**: Industrial process heat or water facility efficiency improvements
- **Actions**:
  1. Map industrial zones near geothermal prospects
  2. Conduct energy audits of potential users
  3. Develop standardized geothermal heat exchange packages
  4. Implement pilot projects with cost-sharing arrangements
  5. Create financing mechanisms (ESCO model, green bonds, etc.)

2.6.3 Phase 3: Electrical Generation (24-60 months)

- **Target**: 100kW-5MW binary cycle plants at best prospects
- **Actions**:
  1. Complete resource confirmation drilling (mitigate exploration risk)
  2. Secure Power Purchase Agreements (PPAs) with REG or industries
  3. Develop financing package (concessional loans, climate funds, equity)
  4. EPC contracting and plant construction
  5. Grid integration and commissioning

2.7 Risk Mitigation Strategies

1. **Exploration Risk**:
   - Start with direct use applications requiring lower flow/temperature certainty
   - Use government-sponsored drilling for initial resource confirmation
   - Implement phased investment tied to resource verification
2. **Market Risk**:
   - Secure off-take agreements before major investments
   - Start with applications serving existing markets (food, fish)
   - Develop cooperative models with end-users
3. **Technical Risk**:
   - Use proven, off-the-shelf equipment for direct use applications
   - Engage experienced geothermal consultants for design
   - Implement monitoring and maintenance protocols
4. **Financial Risk**:
   - Seek concessional financing for exploration and pilot phases
   - Structure projects to generate early cash flow from direct use
   - Utilize public-private partnership (PPP) frameworks

2.8 Economic Impact Assessment

2.8.1 Direct Use Scenario (500kW thermal greenhouse heating):

- **Annual Revenue/Savings**: ~$443,000 (displaced diesel)
- **Job Creation**: 5-10 direct jobs (operations, maintenance, agriculture)

- **Indirect Benefits**:
  - Increased agricultural productivity
  - Reduced food imports
  - Export potential for high-value crops
  - Skills development in geothermal technology

2.8.2 Electrical Generation Scenario (1MW binary plant):

- **Annual Revenue**: ~\$1.19M (electricity sales)
- **Job Creation**: 3-5 direct operations jobs
- **Grid Benefits**:
  - Renewable energy contribution (~7.4GWh/year)
  - Peak load support (geothermal provides baseload)
  - Reduced diesel generator usage
  - Carbon emissions reduction (~4,200 tons CO2/year avoided)

2.9 Kenya Geothermal Best Practices: Lessons for Rwanda

Kenya has emerged as a global leader in geothermal energy development, with over 900 MW of installed capacity as of 2026, primarily from the Olkaria geothermal complex. Rwanda can leverage Kenya's extensive experience to accelerate its own geothermal development, particularly in steam power generation and drilling operations.

2.9.1 Drilling Best Practices from Kenya

1. **Phased Drilling Approach**: Kenya uses a systematic approach starting with slim holes (6-8.5") for exploration, followed by production wells (9.5-12.25") after resource confirmation, reducing wasted expenditure on non-productive wells.
2. **Directional Drilling**: Utilizing directional drilling to target specific fracture zones and steam-rich areas, increasing well productivity by 30-50% compared to vertical wells.
3. **Managed Pressure Drilling (MPD)**: Implementing MPD techniques to handle high-pressure, high-temperature zones safely, reducing non-productive time and improving safety.
4. **Lost Circulation Management**: Developing specialized lost circulation materials (LCM) tailored to volcanic formations, significantly reducing drilling losses in fractured zones.
5. **Real-Time Monitoring**: Using downhole sensors and real-time data transmission to optimize drilling parameters and detect issues early.

2.9.2 Steam Power Generation Best Practices

1. **Steam Field Management**: Kenya employs sophisticated reservoir monitoring including pressure/temperature tracking, tracer testing, and pro-

duction profiling to optimize steam field longevity.

2. **Separator Technology**: Utilizing high-efficiency cyclone separators that achieve >99.5% steam purity, protecting turbines from erosion and corrosion.
3. **Non-Condensable Gas (NCG) Handling**: Implementing NCG removal systems (typically using steam ejectors or compressors) to maintain turbine efficiency and reduce atmospheric emissions.
4. **Modular Power Plant Design**: Standardizing on modular 150-350 MW units that allow for phased expansion and easier maintenance.
5. **Brines Management**: Developing reinjection strategies that maintain reservoir pressure while minimizing surface environmental impact.

2.9.3 Financial and Risk Mitigation Strategies

1. **Risk Sharing Mechanisms**: Kenya has successfully used partial risk guarantees from development partners (World Bank, AfDB) to cover exploration drilling risks.
2. **Local Content Development**: Investing in Kenyan companies and workforce development has reduced costs and increased project sustainability.
3. **Power Purchase Agreement (PPA) Structures**: Using long-term PPAs with escalation clauses tied to inflation rather than fixed prices, providing revenue certainty.
4. **Climate Finance Access**: Leveraging Kenya's success in accessing Green Climate Fund, Clean Technology Fund, and other climate finance streams.

2.9.4 Implementation Recommendations for Shyuha

1. **Establish Kenya-Rwanda Geothermal Knowledge Exchange**: Create formal partnerships with KenGen (Kenya Electricity Generating Company) and GDC (Geothermal Development Company) for technical cooperation.
2. **Adopt Kenya's Drilling Standards**: Implement Kenya's well design and drilling protocols, adapted for Rwanda's specific geological conditions.
3. **Pilot Modular Approach**: Start with smaller (5-10 MW) modular binary cycle plants to demonstrate viability before scaling to larger flash steam plants.
4. **Develop Local Drilling Capacity**: Partner with Kenyan drilling contractors to transfer skills and potentially establish a regional drilling hub.
5. **Implement Reservoir Management Early**: Begin monitoring and modeling efforts from the first wells to optimize field development from inception.

By incorporating these proven Kenyan best practices, Rwanda can significantly reduce technical risks, improve project economics, and accelerate the timeline for commercial geothermal production.

2.10 Conclusion

For the Shyuha project's goal of identifying the highest potential of success and fastest return on investment, **direct thermal applications of geothermal energy—particularly greenhouse heating and aquaculture—represent the most promising near-term opportunities**. These applications offer:

1. **Fastest ROI** (potentially under 1 year for well-designed systems)
2. **Lower Technical Risk** (using established heat exchange technology)
3. **Immediate Economic Benefits** (displacing expensive imported fuels)
4. **Alignment with Development Goals** (food security, job creation, export earnings)
5. **Scalability** (modular systems that can grow with demonstrated success)

While electrical generation from geothermal holds long-term promise for meeting Rwanda's renewable energy targets, the exploration risk, higher capital requirements, and longer payback periods make it a medium- to long-term strategy.

The optimal approach for Shyuha would be to initiate pilot direct use projects that demonstrate viability and create a foundation for eventual electrical generation development as the resource is better characterized and financing mechanisms are established.

2.11 Sources Consulted

1. Rwanda Energy Group - Geothermal Section
2. UNIDO Geothermal Fact Sheet for Rwanda (PDF)
3. Rwanda Dispatch - Geothermal Schemes Article
4. [WATER_PUMPING_ENERGY] - Local analysis of Kigali water pumping energy requirements
5. [SOLAR_ROI_ANALYSIS] - Local analysis of solar ROI for comparison
6. Rwanda energy sector summary
7. Geothermal energy research findings
8. JICA project findings on Rwanda Electricity Development Plan

Note: Financial estimates are based on available public data and industry benchmarks. Site-specific feasibility studies are recommended for accurate project economics.

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3 Geothermal Power Generation Business Plan for Rwanda

3.1 Executive Summary

This business plan outlines a comprehensive strategy for developing geothermal power generation capacity in Rwanda, leveraging the country's significant geothermal resources to meet growing electricity demand, reduce dependence on fossil fuels, and support national development goals. The plan focuses on a phased approach that begins with resource validation and pilot projects, progressing to commercial-scale power plants that will contribute to Rwanda's target of increasing installed capacity from 216 MW to 512 MW by 2024 and achieving 100% electricity access.

3.1.1 Key Opportunity

Rwanda possesses conservative geothermal estimates of up to 100 MW, with some sources suggesting potential exceeding 700 MW. The Northwestern Volcanic Area (Gisenyi, Karisimbi, Kinigi) and Southern Rift-Fault Area (Bugarama) represent the most promising prospect zones for development.

3.1.2 Business Model

The venture will operate as an Independent Power Producer (IPP) developing, owning, and operating geothermal power plants under long-term Power Purchase Agreements (PPAs) with Rwanda Energy Group (REG) or direct sales to industrial customers. Revenue will be primarily generated through electricity sales at regulated tariffs, supplemented by potential carbon credit revenues and ancillary services.

3.1.3 Financial Highlights

- **Initial Investment**: $16.25-$32.5 million for a 5-10 MW binary cycle plant
- **Annual Revenue**: $5.95-$11.9 million (for 5-10 MW plant)
- **Simple Payback Period**: 2.7 years (after exploration risk mitigation)
- **25-Year NPV**: $55-$150 million (8% discount rate)
- **Job Creation**: 3-5 direct operations jobs per MW, plus indirect employment

3.1.4 Implementation Timeline

- **Phase 1 (Q4 2026-Q2 2027)**: Foundation building, resource validation, partnerships
- **Phase 2 (Q3 2027-Q4 2028)**: Pilot implementation, exploratory drilling
- **Phase 3 (2029 and beyond)**: Scale and replicate, commercial plant commissioning

3.2 1. Company Overview

3.2.1 1.1 Vision

To become Rwanda's leading geothermal power producer, providing clean, reliable, and affordable baseload electricity to support the nation's industrialization and sustainable development goals.

3.2.2 1.2 Mission

To develop Rwanda's geothermal resources through innovative technology, strategic partnerships, and environmentally responsible practices, delivering superior returns to stakeholders while advancing national energy security.

3.2.3 1.3 Core Values

- Safety and environmental stewardship
- Technical excellence and innovation
- Partnership and collaboration
- Transparency and integrity
- Long-term value creation

3.2.4 1.4 Legal Structure

The venture will be established as a Rwandan limited liability company, with potential for international investment partners and potential listing on the Rwanda Stock Exchange as it matures.

3.3 2. Market Analysis

3.3.1 2.1 Rwanda's Energy Context

3.3.1.1 Current Situation

- **Installed Capacity**: ~216 MW (as of MININFRA data)
- **Target Capacity by 2024**: 512 MW (requiring ~296 MW of new capacity)
- **Electricity Access**: 51% of households (37% grid, 14% off-grid)
- **Access Goal**: 100% by 2024 (52% on-grid, 48% off-grid)
- **Current Reliance**: Significant dependence on diesel generators and imported electricity

3.3.1.2 Demand Drivers

- **Economic Growth**: Rwanda's Vision 2020 and National Strategy for Transformation (NST1) targeting middle-income status
- **Industrialization**: Made in Rwanda initiative and Kigali Innovation City
- **Urbanization**: Rapid growth of Kigali and secondary cities
- **Regional Integration**: Participation in East African Power Pool (EAPP)
- **Climate Commitments**: 38% emissions reduction pledge by 2030 under Paris Agreement

3.3.2 2.2 Geothermal Resource Assessment

3.3.2.1 Potential Estimates

- **Conservative**: Up to 100 MW (Rwanda Energy Group, African Geothermal Association)
- **Optimistic**: Potential exceeding 700 MW (Rwanda Dispatch)

3.3.2.2 Prospect Zones

1. **Northwestern Volcanic Area**
   - Gisenyi: Hot springs at 70-75°C
   - Karisimbi: Previous drilling (2006) - no viable system found
   - Kinigi: Prospect zone requiring detailed investigation
2. **Southern Rift-Fault Area**
   - Bugarama: Gradient-well work ongoing, primary current focus
3. **Additional Areas**
   - Near Lake Kivu
   - Volcanoes National Park periphery
   - Rubavu Graben
   - Mashyuza

3.3.2.3 Resource Characteristics

- **Temperature Range**: 70-75°C+ at surface manifestations (suitable for binary cycle)
- **Depth to Resource**: To be confirmed through exploration drilling
- **Fluid Chemistry**: Requires assessment for scaling/corrosion potential
- **Permeability**: Fracture-controlled systems expected in volcanic terrain

3.3.3 2.3 Competitive Landscape

3.3.3.1 Current Energy Mix

- Hydropower: ~50% of generation (seasonally variable)
- Diesel: ~30% (high operating costs)
- Methane (Lake Kivu): ~10%
- Solar: Growing but still small percentage
- Imports: Variable from neighboring countries

3.3.3.2 Competitive Advantages of Geothermal

- **Baseload Capacity**: 90-95% capacity factor vs. 20-30% for solar, 40-60% for hydro
- **Predictable Output**: Not weather-dependent
- **Low Operating Costs**: Minimal fuel costs after capital investment
- **Long Lifespan**: 25-30+ years with proper maintenance
- **Small Footprint**: Less land-intensive than solar or hydro per MW

- **Local Resource**: Reduces foreign exchange exposure for fuel imports

3.3.3.3 Market Positioning

The venture will position itself as a provider of clean, reliable baseload power that complements Rwanda's existing hydro resources and reduces dependence on expensive diesel generation. Long-term PPAs will provide revenue certainty while supporting national grid stability.

3.4 3. Technology Description

3.4.1 3.1 Selected Technology: Binary Cycle Organic Rankine Cycle (ORC)

Given the resource temperature range of 70-75°C+ identified at surface manifestations, binary cycle technology is the most appropriate for Rwanda's geothermal resources.

3.4.1.1 How Binary Cycle Works

1. Hot geothermal fluid is brought to surface through production wells
2. Heat is transferred to a secondary working fluid (typically isobutane or similar organic compound) with a lower boiling point than water
3. The secondary fluid vaporizes, driving a turbine connected to an electrical generator
4. The working fluid is condensed and recycled in a closed loop
5. The cooled geothermal fluid is reinjected into the reservoir

3.4.1.2 Advantages for Rwanda

- **Temperature Suitability**: Efficient operation at 70-150°C range
- **Environmental Protection**: Zero emissions, closed-loop system
- **Proven Technology**: Commercial success in similar temperature resources globally
- **Modular Design**: Facilitates phased implementation
- **Grid Compatibility**: Standard electrical output compatible with REG infrastructure

3.4.2 3.2 Plant Components

3.4.2.1 Surface Facilities

- Production and injection wells
- Heat exchangers (primary and secondary)
- Organic Rankine Cycle power block (turbine, generator, condenser)
- Cooling system (air-cooled condensers appropriate for Rwanda's climate)
- Electrical equipment (transformers, switchgear, controls)
- Reinjection system
- Ancillary facilities (control building, storage, workshops)

3.4.2.2 Subsurface Components

- Production wells targeting fracture zones
- Injection wells for reservoir pressure maintenance
- Downhole pumps (if natural insufficient)
- Well casings and cementing for zonal isolation
- Monitoring equipment (temperature, pressure, flow)

3.4.3 3.3 Technology Partners and Best Practices

The venture will leverage Kenya's extensive geothermal experience, adopting proven best practices:

3.4.3.1 Drilling Best Practices (from Kenya)

- **Phased Approach**: Slim holes (6-8.5") for exploration, followed by production wells (9.5-12.25") after confirmation
- **Directional Drilling**: Target specific fracture zones, increasing productivity by 30-50%
- **Managed Pressure Drilling (MPD)**: Handle high-pressure zones safely
- **Lost Circulation Management**: Specialized materials for volcanic formations
- **Real-Time Monitoring**: Downhole sensors for optimization and early issue detection

3.4.3.2 Power Generation Best Practices

- **Steam Field Management**: Pressure/temperature tracking, tracer testing, production profiling
- **High-Efficiency Separators**: >99.5% steam purity protection for turbines
- **NCG Handling**: Steam ejectors or compressors to maintain turbine efficiency
- **Modular Design**: Standardized 150-350 MW units for phased expansion
- **Brines Management**: Reinjection strategies to maintain reservoir pressure

3.5 4. Marketing and Sales Strategy

3.5.1 4.1 Target Customers

Primary: Rwanda Energy Group (REG) through Power Purchase Agreements (PPAs) Secondary: Large industrial customers seeking reliable, clean power (mines, manufacturing) Tertiary: Off-grid or mini-grid applications for remote communities

3.5.2 4.2 Pricing Strategy

- **PPA Tariffs**: Negotiated with REG based on avoided cost of diesel generation
- **Escalation Clauses**: Tied to inflation indices (similar to Kenyan PPAs)
- **Take-or-Pay Provisions**: Ensure revenue certainty
- **Ancillary Services**: Potential revenue from frequency regulation, spinning reserve

3.5.3 4.3 Sales Process

1. **Resource Validation**: Demonstrate measured flow and temperature
2. **Technical Studies**: Grid impact, interconnection studies
3. **PPA Negotiation**: Standardized terms based on REG/RURA frameworks
4. **Financing Securement**: Commitments from lenders and investors
5. **EPC Contracting**: Fixed-price, turnkey construction
6. **Commissioning and Commercial Operation**: Performance testing and handover

3.5.4 4.4 Differentiation Factors

- **Reliability**: Baseload power with >90% availability
- **Clean Energy**: Zero operational emissions supporting climate goals
- **Local Development**: Job creation, technology transfer, economic stimulation
- **Price Stability**: Not subject to international fuel price volatility
- **Long-Term Partnership**: 25-30 year asset life supporting sustained collaboration

3.6 5. Operations Plan

3.6.1 5.1 Plant Operations

- **Staffing**: 3-5 direct operations personnel per MW (shift-based)
- **Maintenance**: Preventive and predictive maintenance programs
- **Monitoring**: SCADA systems for real-time performance tracking
- **Environmental Compliance**: Monitoring of reinjection, surface facilities, emissions
- **Safety Programs**: Comprehensive HSE management systems

3.6.2 5.2 Resource Management

- **Reservoir Monitoring**: Pressure, temperature, flow rates, chemical composition
- **Reinjection Strategy**: Maintain reservoir pressure and sustainability
- **Make-up Well Planning**: Reserve for reservoir decline compensation
- **Production Optimization**: Balance extraction with reservoir capacity

3.6.3 5.3 Maintenance Philosophy

- **Preventive Maintenance**: Regular scheduled maintenance based on OEM recommendations
- **Predictive Maintenance**: Condition monitoring, vibration analysis, oil analysis
- **Corrective Maintenance**: Rapid response to failures with spare parts inventory
- **Turnaround Planning**: Scheduled major maintenance every 3-5 years

3.6.4 5.4 Quality and Standards

- **International Standards**: Adherence to IEC, ISO, and relevant geothermal industry standards
- **Local Compliance**: RURA, REMA, and other Rwandan regulatory requirements
- **Best Practices**: Continuous improvement through benchmarking with global leaders
- **Certifications**: Pursue relevant quality, environmental, and safety certifications

3.7 6. Management Team

3.7.1 6.1 Organizational Structure

```
Board of Directors
    |
Chief Executive Officer (CEO)
    |
    |----------------|----------------|----------------|
Chief Technical Officer (CTO)  Chief Financial Officer (CFO)  Chief Operations Officer (COO)
    |                      |                       |
Geothermal Team         Finance Team         Operations Team
Drilling/Reservoir      Accounting/Reporting Plant Management
Power Plant             Treasury/Financing   Maintenance
Environmental           Investor Relations   Health & Safety
Community Relations                             Grid Integration
```

3.7.2 6.2 Key Personnel Requirements

- **CEO**: Energy sector experience, preferably with geothermal or power development background
- **CTO**: Geothermal reservoir engineering, drilling, or power plant experience (Kenyan expertise preferred initially)
- **CFO**: Project finance, infrastructure funding, or energy sector finance experience

- **COO**: Power plant operations, preferably geothermal or similar baseload experience
- **Geothermal Specialists**: Reservoir engineers, geologists, drilling engineers
- **Technical Team**: Mechanical, electrical, and control engineers
- **Local Team**: Rwandan professionals for community relations, permitting, and local content

3.7.3 6.3 Advisory Board

- **Technical Advisors**: KenGen, GDC, or other geothermal experts
- **Financial Advisors**: Infrastructure finance specialists
- **Legal Advisors**: Energy law, PPP specialists, local counsel
- **Environmental Advisors**: ESIA experts, sustainability specialists

3.7.4 6.4 Human Resources Strategy

- **Knowledge Transfer**: Formal programs with Kenyan partners
- **Local Capacity Building**: Training programs, university partnerships
- **Performance Management**: Clear KPIs, regular reviews, competitive compensation
- **Succession Planning**: Identification and development of future leaders
- **Culture**: Safety-first, collaborative, innovative, results-oriented

3.8 7. Financial Plan

3.8.1 7.1 Capital Requirements

3.8.1.1 Phase 1: Foundation and Validation (Q4 2026-Q2 2027)

- **Technical Studies**: $500,000-$1,000,000
  - Surface geoscience, geochemistry, preliminary modeling
- **Partnership Development**: $200,000-$400,000
  - Technical exchange programs, MoUs with Kenyan partners
- **Capacity Building**: $300,000-$500,000
  - Training, workshops, knowledge transfer
- **Pilot Design**: $400,000-$600,000
  - Direct-use system design for greenhouse/aquaculture
- **Total Phase 1**: $1.4-$2.5 million

3.8.1.2 Phase 2: Pilot Implementation and Exploration (Q3 2027-Q4 2028)

- **Direct-Use Pilot**: $250,000-$750,000
  - 100-500kW thermal greenhouse heating system
- **Exploratory Drilling**: $3,000,000-$6,000,000
  - 2-3 slim holes + 1-2 production wells at Kinigi/Gisenyi
- **Reservoir Monitoring**: $500,000-$800,000

- Downhole equipment, surface monitoring systems
- **Business Model Development**: $200,000-$400,000
  - Scale-up planning based on pilot results
- **Total Phase 2**: $4.15-$7.95 million

3.8.1.3 Phase 3: Commercial Scale (2029 and beyond)

- **5 MW Binary Cycle Plant**: $12.5-$20.0 million
  - EPC, grid connection, contingency
- **10 MW Binary Cycle Plant**: $25.0-$40.0 million
  - EPC, grid connection, contingency
- **Total Phase 3 (5-10 MW)**: $12.5-$40.0 million

3.8.2 7.2 Revenue Projections

3.8.2.1 Assumptions for 5-10 MW Binary Cycle Plant

- **Capacity Factor**: 85% (conservative for baseload geothermal)
- **Degradation**: 0.5% per year
- **Electricity Tariff**: $0.16/kWh (Rwanda commercial/industrial rate)
- **Plant Lifetime**: 25 years
- **Availability**: 90-95% (including planned maintenance)

3.8.2.2 Annual Generation and Revenue

Plant Size	Gross Annual Generation	Annual Revenue (Year 1)	Notes
5 MW	37.2 GWh	$5.95 million	85% CF, $0.16/kWh
10 MW	74.4 GWh	$11.9 million	85% CF, $0.16/kWh

Note: Revenue assumes full PPA contract. Early years may have lower output during ramp-up.

3.8.3 7.3 Operating Expenses

3.8.3.1 Annual O&M Costs

- **Fixed O&M**: 1-3% of CAPEX annually
- **Variable O&M**: Minimal (no fuel costs)
- **Parasitic Load**: 10-15% of gross generation (used internally)
- **Insurance**: Standard power plant coverage
- **Property Taxes**: As per Rwandan regulations
- **Administrative**: Management, reporting, compliance

3.8.3.2 Example O&M for 5 MW Plant ($16.25M CAPEX)

- **Fixed O&M**: $162,500-$487,500/year (1-3%)
- **Variable/Other**: Estimated $100,000-$200,000/year
- **Total Annual O&M**: $262,500-$687,500/year

3.8.4 7.4 Profitability Analysis

3.8.4.1 5 MW Plant Example ($16.25M CAPEX)

Metric	Calculation	Value
Annual Revenue	37.2 GWh × $0.16/kWh	$5.95 million
Annual O&M	2% of CAPEX	$0.325 million
EBITDA	Revenue - O&M	$5.625 million
EBITDA Margin	EBITDA/Revenue	94.5%
Simple Payback	CAPEX/EBITDA	2.9 years
25-Year NPV (8%)	DCF of cash flows	~$55 million

3.8.4.2 10 MW Plant Example ($32.5M CAPEX)

Metric	Calculation	Value
Annual Revenue	74.4 GWh × $0.16/kWh	$11.9 million
Annual O&M	2% of CAPEX	$0.65 million
EBITDA	Revenue - O&M	$11.25 million
EBITDA Margin	EBITDA/Revenue	94.5%
Simple Payback	CAPEX/EBITDA	2.9 years
25-Year NPV (8%)	DCF of cash flows	~$110 million

Note: These estimates exclude exploration/drilling risk costs, which are addressed in Phase 1-2.

3.8.5 7.5 Funding Sources

3.8.5.1 Equity

- **Founders' Capital**: Initial seed funding
- **Strategic Investors**: Energy companies, infrastructure funds
- **Development Finance Institutions**: DFIs taking equity stakes
- **Local Investors**: Rwandan high-net-worth individuals, institutions
- **Employee Ownership**: Potential ESOP for key personnel

3.8.5.2 Debt

- **Concessional Loans**: World Bank, AfDB, KfW, etc. for exploration and early stages

- **Commercial Debt**: Local and international banks for construction phase
- **Green Bonds**: Climate-focused instruments for environmentally beneficial projects
- **Vendor Financing**: EPC contractor deferred payments

3.8.5.3 Grants and Technical Assistance

- **Climate Finance**: Green Climate Fund, Clean Technology Fund
- **Technical Assistance**: UNDP, UNIDO, GIZ for capacity building
- **Research Grants**: Academic and international organization funding
- **Risk Guarantees**: Partial risk guarantees from DFIs for exploration drilling

3.8.6 7.6 Financial Sensitivities

3.8.6.1 Key Variables Impacting Project Economics

1. **Resource Temperature/Flow**: Directly impacts plant size and efficiency
2. **Drilling Success Rate**: Affects capital efficiency and timeline
3. **Electricity Tariff**: Primary revenue driver
4. **Capital Costs**: EPC pricing, contingencies, currency fluctuations
5. **Capacity Factor**: Plant availability and grid constraints
6. **O&M Costs**: Operational efficiency and maintenance effectiveness
7. **Exchange Rates**: For imported equipment and USD-denominated debt
8. **Tax Incentives**: Potential tax holidays or reduced rates for renewable energy

3.9 8. Implementation Plan

3.9.1 8.1 Phased Approach

3.9.1.1 Phase 1: Foundation Building (Q4 2026-Q2 2027) **Objective**: Validate resource potential, establish partnerships, prepare for drilling **Duration**: 6 months **Key Activities**: - Complete geothermal direct-use profitability analysis - Conduct detailed temperature and flow rate measurements at Gisenyi and Bugarama - Engage with agricultural cooperatives and fish farmers near prospect zones - Initiate Kenya-Rwanda geothermal technical exchange program with KenGen and GDC - Begin capacity building for Rwandan professionals through Kenyan training initiatives - Design pilot direct-use system (50-200kW thermal) for greenhouse heating or aquaculture - Finalize solar ROI models for various scales (100kW-10MW) targeting Kigali City - Identify potential sites for pilot solar installations - Engage with MININFRA and REG on net metering and interconnection policies - Develop financing concepts for solar PPAs and green bonds - Conduct preliminary assessment of Kigali's water pumping energy requirements - Identify key pumping stations for energy audits - Research VFDs, high-efficiency pumps, and solar PV integration options

Deliverables: - Validated geothermal resource data - Established technical partnerships - Capacity building program underway - Pilot direct-use system design completed - Financial models for geothermal, solar, and water pumping efficiency

3.9.1.2 Phase 2: Pilot Implementation (Q3 2027-Q4 2028)

Objective: Demonstrate viability through pilots, commence exploration drilling **Duration**: 6 months (Note: Timeline appears compressed in source doc - likely 18 months) **Key Activities**: - Launch 100kW thermal greenhouse heating pilot at Gisenyi - Install monitoring systems to track performance and ROI - Develop business model for scale-up based on pilot results - Commence exploratory drilling at Kinigi/Gisenyi using Kenyan best practices and technical partners - Implement reservoir management monitoring from initial wells - Explore aquaculture facility heating pilots alongside greenhouse projects - Deploy 500kW solar PV system on government building - Implement net metering arrangement with REG - Begin commercial/industrial rooftop solar outreach program - Develop standardized solar PPA templates for large consumers - Conduct grid impact studies for utility-scale solar integration - Conduct water-pumping energy audit at Karenge treatment plant - Implement VFDs on major pumping stations - Pilot solar PV integration for daytime pumping operations - Conduct leak detection and repair initiative in distribution network - Explore geothermal-assisted heat pumps for water facilities

Deliverables: - Operational direct-use pilot with verified ROI - Commenced exploratory drilling program - Reservoir monitoring system operational - Solar PV system generating electricity - Water pumping efficiency improvements demonstrated - Financing mechanisms identified for scale-up

3.9.1.3 Phase 3: Scale and Replicate (2029 and Beyond)

Objective: Commercial geothermal power generation and nationwide expansion **Duration**: Ongoing **Key Activities**: - Expand successful pilots to additional sites (Bugarama, Kinigi, etc.) - Develop 1-5MW geothermal binary cycle plants at confirmed resources - Create franchise model for geothermal direct-use systems nationwide - Establish joint ventures with Kenyan partners for larger geothermal power projects (>5MW) - Develop local drilling capacity through partnerships with Kenyan contractors - Implement advanced reservoir management for steam field optimization - Scale to 1-5MW commercial/industrial solar installations - Develop utility-scale (5MW+) grid-connected solar projects with storage - Establish local solar panel assembly and maintenance capabilities - Create solar financing vehicle for residential and small business adoption - Integrate solar with agricultural processing and cold chain facilities - Implement system-wide pump and motor efficiency upgrades - Expand solar PV pumping to additional treatment plants and booster stations - Implement comprehensive pressure management and leak detection programs - Explore hydropower recovery from pressure reduction valves - Develop integrated renewable energy pumping systems (solar + grid + storage)

Deliverables: - Commercial geothermal power plants operational - Nationwide direct-use franchise model - Local drilling capacity established - Integrated renewable energy solutions deployed - Measurable impact on energy access and renewable targets

3.9.2 8.2 Critical Path Items

1. **Resource Confirmation**: Successful exploration drilling with measurable flow/temperature
2. **Partnership Agreements**: Formal technical and commercial agreements with Kenyan partners
3. **Financing Closure**: Commitments from equity and debt providers
4. **Regulatory Approvals**: Environmental, construction, and generation licenses
5. **PPA Execution**: Signed power purchase agreement with REG
6. **EPC Contracting**: Fixed-price turnkey contract for power plant
7. **Grid Connection**: Approved interconnection study and transmission upgrades if needed
8. **Construction Completion**: On-time, on-budget power plant construction
9. **Commissioning**: Successful performance testing and handover
10. **Commercial Operation**: Sustained generation meeting PPA requirements

3.9.3 8.3 Milestone Schedule

Milestone	Target Date	Description
Phase 1 Completion	Q2 2027	Foundation building completed, partnerships established
Exploratory Drilling Start	Q3 2027	Commence drilling at Kinigi/Gisenyi
Resource Confirmation	Q2 2028	Measured flow and temperature confirmed
PPA Signed	Q3 2028	Power Purchase Agreement with REG executed
Financial Close	Q4 2028	Debt and equity financing secured
EPC Contract Award	Q1 2029	Engineering, Procurement, Construction contract awarded
Construction Start	Q2 2029	Groundbreaking for geothermal power plant

Milestone	Target Date	Description
Mechanical Completion	Q4 2029	Plant construction completed
Commissioning Start	Q1 2030	Systems testing and initialization
Commercial Operation Date (COD)	Q2 2030	Full commercial operation under PPA
Phase 2 Completion	Q4 2028	Pilot implementation completed
Phase 3 Milestone 1	2031	First 1-5MW geothermal plant commissioned
Phase 3 Milestone 2	2033	Joint ventures established with Kenyan partners
Phase 3 Milestone 3	2035	Local drilling capacity operational

3.10 9. Risk Management

The detailed risk register is maintained in `RISK_REGISTER.md`. Key risks and mitigations include:

3.10.1 9.1 Critical Risks (Score 20-25)

3.10.1.1 R1: No Commercially Viable Resource Found

- **Likelihood**: 5 (Near certain >80%)
- **Impact**: 5 (Threatens venture existence)
- **Score**: 25 Critical
- **Mitigation**: Phase 1 structured as falsification test with pre-registered numeric kill criteria; Phase 3 capital contingent on gate; do not commit generation capital before measured flow and temperature exist
- **Residual**: 20 Critical (still high due to inherent exploration risk)

3.10.1.2 R2: Duplication with EDCL Efforts

- **Likelihood**: 4 (Likely 60-80%)
- **Impact**: 5 (Major — delays or re-scopes a phase)
- **Score**: 20 Critical
- **Mitigation**: Position explicitly as commercialisation layer, not explorer; take proven resource under resource-access agreement; confirm relationship in writing before approaching government
- **Residual**: 8 Medium

3.10.2 9.2 High Risks (Score 12-16)

3.10.2.1 R3: GRMF Ineligibility

- **Likelihood**: 4 (Likely)
- **Impact**: 4 (Major)
- **Score**: 16 High
- **Mitigation**: Recruit technical partner with East African geothermal delivery record; apply as partner to EDCL's programme
- **Residual**: 8 Medium

3.10.2.2 R4: No Direct-Use Customer

- **Likelihood**: 4 (Likely)
- **Impact**: 4 (Major)
- **Score**: 16 High
- **Mitigation**: Audit prospective customers' current fuel use before pilot design; reframe as grant-funded demonstration if needed
- **Residual**: 12 High

3.10.2.3 R5: Unknown Generation Tariff

- **Likelihood**: 4 (Likely)
- **Impact**: 4 (Major)
- **Score**: 16 High
- **Mitigation**: Treat tariff as model output (breakeven) until RURA REFIT level or tender result confirmed
- **Residual**: 8 Medium

3.10.2.4 R6: Drilling Costs Exceed Budget

- **Likelihood**: 4 (Likely)
- **Impact**: 3 (Moderate)
- **Score**: 12 High
- **Mitigation**: Benchmark from Kenyan actuals with explicit contingency; fixed-price drilling contracts; partial risk guarantee cover
- **Residual**: 6 Medium

3.10.2.5 R7: Currency Movement

- **Likelihood**: 3 (Possible)
- **Impact**: 4 (Major)
- **Score**: 12 High
- **Mitigation**: Establish PPA currency at term-sheet; local-currency financing for RWF-denominated costs; state FX assumption explicitly
- **Residual**: 6 Medium

3.10.2.6 R8: Land Access Constraints

- **Likelihood**: 3 (Possible)
- **Impact**: 4 (Major)
- **Score**: 12 High
- **Mitigation**: Confirm tenure and protected-area status early; engage REMA; note northern prospects near Volcanoes National Park
- **Residual**: 6 Medium

3.10.2.7 R9: Permitting Delays

- **Likelihood**: 4 (Likely)
- **Impact**: 3 (Moderate)
- **Score**: 12 High
- **Mitigation**: Early engagement with REMA and local authorities; sequence licence application ahead of construction decision
- **Residual**: 6 Medium

3.10.3 9.3 Risk Management Approach

3.10.3.1 Risk Monitoring

- **Quarterly Reviews**: Board-level review of R2-R5, R7 with updates
- **Phase Gates**: Re-rate R1 and R6 based on measured results at Phase 1 completion
- **Annual Full Review**: Complete re-rate aligned with Rwanda's national planning cycle
- **Ad-hoc Reviews**: On material policy, financing, or technical changes

3.10.3.2 Risk Mitigation Principles

1. **Exploration Risk**: Start with direct-use applications requiring lower flow/temperature certainty; use government-sponsored drilling for initial confirmation; implement phased investment tied to resource verification
2. **Market Risk**: Secure off-take agreements before major investments; start with applications serving existing markets (food, fish); develop cooperative models with end-users
3. **Technical Risk**: Use proven, off-the-shelf equipment for direct-use applications; engage experienced geothermal consultants for design; implement monitoring and maintenance protocols
4. **Financial Risk**: Seek concessional financing for exploration and pilot phases; structure projects to generate early cash flow from direct use; utilize public-private partnership (PPP) frameworks

3.11 10. Economic and Social Impact

3.11.1 10.1 National Energy Contribution

3.11.1.1 Capacity Contribution

- **5 MW Plant**: ~1% of Rwanda's 2024 target capacity (512 MW)
- **10 MW Plant**: ~2% of Rwanda's 2024 target capacity
- **Multiple Plants**: Potential for 10-20% of target capacity with successful exploration

3.11.1.2 Generation Characteristics

- **Baseload Power**: 90-95% availability complements variable hydro and solar
- **Peak Load Support**: Can provide dispatchable power during peak periods
- **Grid Stability**: Frequency regulation and spinning reserve capabilities
- **Reduced Diesel Displacement**: Direct reduction in expensive diesel generator usage

3.11.1.3 Energy Security Benefits

- **Import Reduction**: Decreased reliance on imported electricity and diesel fuel
- **Foreign Exchange Savings**: Lower fuel import bills
- **Price Stability**: Insulated from international fuel price volatility
- **Domestic Resource**: Utilizes indigenous energy source

3.11.2 10.2 Employment and Skills Development

3.11.2.1 Direct Employment

- **Construction Phase**: 50-150 jobs (EPC contractors and subcontractors)
- **Operations Phase**: 3-5 direct jobs per MW (shift-based operations, maintenance, supervision)
- **Indirect Employment**: 2-4 indirect jobs per direct job (suppliers, services, induced)

3.11.2.2 Skills Development

- **Technical Training**: Geothermal reservoir engineering, drilling, power plant operations
- **Local Content Development**: Partnerships with University of Rwanda, IPRC for curriculum development
- **Knowledge Transfer**: Formal programs with Kenyan experts (KenGen, GDC)
- **Certification Programs**: Operator and technician certification pathways

- **Entrepreneurship**: Spin-off opportunities in geothermal services, maintenance, consulting

3.11.3 10.3 Environmental Benefits

3.11.3.1 Emissions Reduction

- **CO2 Avoidance**: ~4,200 tons CO2/year avoided per 1MW plant (vs. diesel)
- **Other Pollutants**: Near-zero SOx, NOx, particulates, mercury compared to fossil fuels
- **Lifecycle Emissions**: Minimal compared to fossil fuel alternatives

3.11.3.2 Resource Stewardship

- **Reinjection**: Sustainable reservoir management through fluid return
- **Surface Footprint**: ~0.5-1 acre per MW (much less than solar or hydro)
- **Water Use**: Minimal consumptive use (binary cycle is closed-loop)
- **Visual Impact**: Low-profile facilities compatible with tourism areas
- **Seismic Monitoring**: Induced seismicity monitoring and management protocols

3.11.4 10.4 Alignment with National Goals

3.11.4.1 Vision 2020 and NST1

- **Economic Transformation**: Supports industrialization through reliable power
- **Infrastructure Development**: Contributes to energy infrastructure goals
- **Private Sector Engagement**: PPP model encourages private investment
- **Regional Integration**: Supports East African Power Pool objectives

3.11.4.2 Climate Commitments

- **NDC Contribution**: Direct contribution to 38% emissions reduction pledge
- **Clean Energy Transition**: Advances renewable energy share in generation mix
- **Sustainable Development**: Supports SDG 7 (Affordable and Clean Energy)

3.11.4.3 Social Development Goals

- **Energy Access**: Enables expansion of grid to underserved areas
- **Job Creation**: Direct and indirect employment opportunities

- **Skills Development**: Technical capacity building for youth and professionals
- **Local Content**: Maximizes Rwandan participation in value chain

3.12 11. Conclusions and Recommendations

3.12.1 11.1 Investment Thesis

The geothermal power generation opportunity in Rwanda presents a compelling investment case based on:

1. **Strong National Need**: Clear electricity supply gap and access targets
2. **Significant Resource Potential**: Conservative estimates of 100MW+, with higher estimates suggesting 700MW+
3. **Appropriate Technology**: Binary cycle well-suited to 70-75°C+ resource temperatures
4. **Proven Economic Model**: Attractive returns with 2.7-year payback and 25-year NPV of $55-$150 million for 5-10MW plants
5. **Strategic Partnerships**: Opportunity to leverage Kenya's extensive geothermal experience
6. **National Priority**: Alignment with Vision 2020, NST1, and climate commitments
7. **Risk Mitigation**: Phased approach that validates resource before major capital commitment

3.12.2 11.2 Recommended Next Steps

3.12.2.1 Immediate Actions (0-3 months)

1. **Finalize Team**: Recruit key leadership (CEO, CTO with geothermal experience)
2. **Stakeholder Engagement**: Formal introductions to MININFRA, REG, REMA, RDB
3. **Partnership Outreach**: Initiate discussions with KenGen and GDC for technical cooperation
4. **Preliminary Budget**: Refine Phase 1 budget and identify potential funding sources
5. **Work Planning**: Develop detailed work plan for Phase 1 foundation activities

3.12.2.2 Short-Term Actions (3-6 months)

1. **Legal Establishment**: Register company, establish governance structure
2. **Initial Technical Work**: Commission desktop geothermal study of prospect zones
3. **Partnership Agreements**: Sign MoUs with Kenyan technical partners
4. **Capacity Building Plan**: Develop training program with Kenyan institutions

5. **Financial Planning**: Prepare preliminary financial model and funding strategy

3.12.2.3 Medium-Term Actions (6-12 months)

1. **Field Campaign**: Conduct temperature and flow measurements at Gisenyi and Bugarama
2. **Stakeholder Workshops**: Engage local communities and prospective direct-use customers
3. **Drilling Planning**: Prepare exploratory drilling program based on surface results
4. **Funding Applications**: Submit proposals to concessional finance sources
5. **Regulatory Engagement**: Begin preliminary discussions with RURA and REMA on requirements

3.12.3 11.3 Final Statement

Rwanda's geothermal resources offer a strategic opportunity to develop clean, reliable baseload power that can significantly advance the nation's energy security, economic development, and climate goals. While exploration risk remains a key challenge, the phased approach outlined in this business plan—beginning with direct-use applications and progressing to power generation as the resource is better characterized—provides a prudent path forward.

By leveraging Kenya's extensive geothermal experience, implementing proven best practices, and maintaining a disciplined, milestone-driven approach, Rwanda can successfully develop its geothermal potential and establish a sustainable source of clean energy for decades to come. The Shyuha project is well-positioned to lead this effort, creating value for investors while delivering tangible benefits to the Rwandan people and economy.

This business plan synthesizes research from SUMMARY.md, HOT_SPRINGS_ENERGY_PROFITABILITY. PROJECT_TIMELINE.md, RISK_REGISTER.md, and other documents in the Shyuha repository. Financial estimates are based on available public data and industry benchmarks. Site-specific feasibility studies are recommended for accurate project economics.

Last Updated: September 2026

4 Shyuha Project: Implementation Timeline and Activities

This document consolidates the implementation timeline and specific activities for the Shyuha project, drawing from all research documents in the repository.

It provides a chronological roadmap for advancing Rwanda's renewable energy initiatives with a focus on geothermal direct-use applications, solar power optimization, and water pumping efficiency.

4.1 Overview

The Shyuha project follows a phased approach designed to minimize risk, demonstrate viability early, and build foundations for larger-scale implementation. The timeline spans from Q4 2026 through 2029 and beyond, with overlapping activities across the three pillars:

1. **Geothermal Direct-Use Applications** (Near-Term Focus)
2. **Solar Power ROI Optimization** (Mid-Term)
3. **Water-Pumping Energy Efficiency** (Cross-Cutting)

4.2 Detailed Timeline

4.2.1 Phase 1: Foundation Building (Q4 2026 - Q2 2027)

Duration: 6 months

4.2.1.1 Geothermal Direct-Use Applications

- Complete geothermal direct-use profitability analysis (HOT_SPRINGS_ENERGY_PROFITABILITY)
- Conduct detailed temperature and flow rate measurements at Gisenyi and Bugarama hot springs
- Engage with agricultural cooperatives and fish farmers near prospect zones
- Initiate Kenya-Rwanda geothermal technical exchange program with KenGen and GDC
- Begin capacity building for Rwandan professionals through Kenyan training initiatives
- Design pilot direct-use system (50-200kW thermal) for greenhouse heating or aquaculture

4.2.1.2 Solar Power Optimization

- Finalize solar ROI models for various scales (100kW-10MW) targeting Kigali City
- Identify potential sites for pilot solar installations on government buildings, schools, and hospitals
- Engage with MININFRA and REG on net metering and interconnection policies
- Develop financing concepts for solar PPAs and green bonds

4.2.1.3 Water-Pumping Energy Efficiency

- Conduct preliminary assessment of Kigali's water pumping energy requirements

- Identify key pumping stations for energy audits
- Research VFDs, high-efficiency pumps, and solar PV integration options
- Map potential gravity-fed options from higher elevation sources

4.2.2 Phase 2: Pilot Implementation (Q3 2027 - Q4 2028)

Duration: 6 months

4.2.2.1 Geothermal Direct-Use Applications

- Launch 100kW thermal greenhouse heating pilot at Gisenyi
- Install monitoring systems to track performance and ROI
- Develop business model for scale-up based on pilot results
- Commence exploratory drilling at Kinigi/Gisenyi using Kenyan best practices and technical partners
- Implement reservoir management monitoring from initial wells
- Explore aquaculture facility heating pilots alongside greenhouse projects

4.2.2.2 Solar Power Optimization

- Deploy 500kW solar PV system on government building
- Implement net metering arrangement with REG
- Begin commercial/industrial rooftop solar outreach program
- Develop standardized solar PPA templates for large consumers
- Conduct grid impact studies for utility-scale solar integration

4.2.2.3 Water-Pumping Energy Efficiency

- Conduct water-pumping energy audit at Karenge treatment plant
- Implement VFDs on major pumping stations
- Pilot solar PV integration for daytime pumping operations
- Conduct leak detection and repair initiative in distribution network
- Explore geothermal-assisted heat pumps for water facilities

4.2.3 Phase 3: Scale and Replicate (2029 and Beyond)

Ongoing

4.2.3.1 Geothermal Direct-Use Applications

- Expand successful pilots to additional sites (Bugarama, Kinigi, etc.)
- Develop 1-5MW geothermal binary cycle plants at confirmed resources
- Create franchise model for geothermal direct-use systems nationwide
- Establish joint ventures with Kenyan partners for larger geothermal power projects (>5MW)
- Develop local drilling capacity through partnerships with Kenyan contractors
- Implement advanced reservoir management for steam field optimization

4.2.3.2 Solar Power Optimization

- Scale to 1-5MW commercial/industrial solar installations
- Develop utility-scale (5MW+) grid-connected solar projects with storage
- Establish local solar panel assembly and maintenance capabilities
- Create solar financing vehicle for residential and small business adoption
- Integrate solar with agricultural processing and cold chain facilities

4.2.3.3 Water-Pumping Energy Efficiency

- Implement system-wide pump and motor efficiency upgrades
- Expand solar PV pumping to additional treatment plants and booster stations
- Implement comprehensive pressure management and leak detection programs
- Explore hydropower recovery from pressure reduction valves
- Develop integrated renewable energy pumping systems (solar + grid + storage)

4.3 Cross-Cutting Activities (Throughout All Phases)

4.3.1 Knowledge Management

- Maintain and update research documents in this repository
- Publish findings and lessons learned in accessible formats
- Host annual Rwanda Geothermal and Renewable Energy Forum
- Develop training curriculum for technicians and professionals
- Create open-access datasets for researchers and investors

4.3.2 Partnership Development

- Maintain engagement with MININFRA, REG, RDB, REMA
- Develop relationships with DFIs (World Bank, AfDB, FMO) for financing
- Engage with private sector investors and developers
- Foster academic collaborations with University of Rwanda and IPRC
- Participate in regional initiatives (EAAPP, Nile Basin Initiative, etc.)

4.3.3 Monitoring and Evaluation

- Track key impact metrics:
  - Energy impact: MWh of clean energy generated/displaced
  - Economic impact: Jobs created, diesel imports reduced, export earnings increased
  - Environmental impact: Tons of CO2 emissions avoided
  - Social impact: Improved energy access, enhanced agricultural productivity
  - Knowledge impact: Stakeholders served, datasets downloaded, citations in policy documents

- Conduct annual reviews and adjust strategy based on results
- Report progress to stakeholders and funding partners

4.4 Risk Management Timeline

4.4.1 Exploration Risk Mitigation

- **Q4 2026-Q2 2027**: Start with direct-use applications requiring lower flow/temperature certainty
- **Q3 2027-Q4 2028**: Use government-sponsored drilling for initial resource confirmation
- **Ongoing**: Implement phased investment tied to resource verification

4.4.2 Market Risk Mitigation

- **Q4 2026-Q2 2027**: Begin securing off-take agreements before major investments
- **Q3 2027-Q4 2028**: Start with applications serving existing markets (food, fish)
- **Ongoing**: Develop cooperative models with end-users

4.4.3 Technical Risk Mitigation

- **Q4 2026-Q2 2027**: Use proven, off-the-shelf equipment for direct-use applications
- **Q3 2027-Q4 2028**: Engage experienced geothermal consultants for design
- **Ongoing**: Implement monitoring and maintenance protocols

4.4.4 Financial Risk Mitigation

- **Q4 2026-Q2 2027**: Seek concessional financing for exploration and pilot phases
- **Q3 2027-Q4 2028**: Structure projects to generate early cash flow from direct use
- **Ongoing**: Utilize public-private partnership (PPP) frameworks

4.5 Success Criteria and Exit Conditions

4.5.1 Phase 1 Completion (End Q2 2027)

- Geothermal direct-use profitability analysis completed and validated
- Kenya-Rwanda technical exchange program established with active participation
- Preliminary partnerships formed with agribusiness cooperatives and municipal utilities
- Solar ROI models finalized for all target scales
- Baseline water pumping energy audit methodology established

4.5.2 Phase 2 Completion (End Q4 2028)

- 100kW thermal greenhouse heating pilot operational with verified ROI
- 500kW solar PV system installed and generating electricity
- Water pumping energy audit completed at Karenge treatment plant
- Exploratory drilling commenced at Kinigi/Gisenyi using Kenyan best practices
- Financing mechanisms identified for scale-up phases

4.5.3 Phase 3 Milestones (2029+)

- Successful expansion of pilot projects to additional sites
- Commissioning of 1-5MW geothermal binary cycle plants
- Establishment of franchise model for geothermal direct-use systems
- Joint ventures established with Kenyan partners for larger projects
- Measurable impact on Rwanda's energy access and renewable energy targets

4.6 Dependencies and Critical Path

4.6.1 Critical Dependencies

1. **Government Engagement**: Alignment with MININFRA and REG policies and timelines
2. **Financing Availability**: Access to concessional loans, climate funds, and private investment
3. **Technical Partnerships**: Ongoing collaboration with Kenyan geothermal experts
4. **Local Capacity**: Development of Rwandan technical and managerial expertise
5. **Land Access**: Secure access to geothermal prospect zones and solar installation sites

4.6.2 Risk Factors and Mitigation

- **Permitting Delays**: Early engagement with REMA and local authorities
- **Financing Gaps**: Diversified funding sources including grants, concessional loans, and equity
- **Technical Challenges**: Phased approach with proven technologies first
- **Market Acceptance**: Pilot demonstrations and off-take agreements
- **Currency Fluctuations**: Hedging strategies and local currency financing where possible

4.7 Review and Update Cycle

This timeline should be reviewed and updated: - Quarterly: Progress tracking and minor adjustments - Biannually: Significant revisions based on pilot results and changing conditions - Annually: Comprehensive review aligned with

Rwanda's national planning cycles - Ad-hoc: Major changes in policy, financing, or technical circumstances

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Timeline last updated: September 2026 Based on research from: HOT_SPRINGS_ENERGY_PROFITABILITY, SOLAR_ROI_ANALYSIS, WATER_PUMPING_ENERGY, SUMMARY, JICA, MININFRA

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5 JICA Project Findings: Rwanda Electricity Development Plan

As of recent research (September 2026):

Project: Project for Preparation of Electricity Development Plan for Rwanda
Agency: Japan International Cooperation Agency (JICA)

Objectives: - Formulate Master Plan for power generation and transmission expansion (15-year horizon) - Develop Master Plan for Sustainable Geothermal Energy Development
- Reduce fossil-fuel dependence and stabilize electricity supply

Current Status: Monitoring results not available (per JICA project listing)

Context: Rwanda currently has no commercial geothermal production despite significant potential (estimated up to 100 MW). Exploration continues with surface studies in Kinigi, Gisenyi, and Bugarama, while Karisimbi drilling found no viable resource. Test sites are planned at Kinigi and Karisimbi for future development.

This JICA-supported initiative aims to create the strategic framework necessary for developing Rwanda's geothermal resources as part of its broader energy diversification strategy outlined in the Energy Sector Strategic Plan.

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6 Summary of MININFRA Energy Sector Information

6.1 Rwanda's Energy Mix

- **Hydropower**: Approximately 47% of installed capacity
  - Nyabarongo I: 28 MW
  - Nyabarongo II: 43.5 MW

  - Rusizi III: 145 MW
  - Rusumo Falls: 80 MW
- **Solar Power**:
  - Mount Jali: 250 kWp
  - Rwamagana: 8.3 MW
  - Nasho: 3.3 MW
  - 30 MW storage plant under study
- **Geothermal Prospects**:
  - Short-term potential: >20 MW
  - Long-term potential: Over 300 MW
  - Test sites planned at Kinigi and Karisimbi
- **Lake Kivu Methane**:
  - Potential: ~350 MW
  - Current production: KivuWatt delivers ~26.4 MW
- **Peat-to-Power Plants**:
  - Gishoma: 15 MW
  - Hakan project: 80 MW

6.2 Energy Access

- **Current Access**: 51% of households
  - Grid connection: 37%
  - Off-grid: 14%
- **Target**: Universal coverage by 2024

6.3 National Energy Consumption

- **Biomass**: Supplies approximately 85% of national energy use

6.4 Policy Framework

Rwanda's energy sector is guided by: - Electricity Law - Rwanda Energy Policy - Biomass Energy Strategy - Energy Sector Strategic Plan

Coordination occurs through the E-SWAP secretariat and regular Joint Sector Reviews.

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7 Shyuha Project: Geothermal Power Generation Business Plan

This repository contains the business plan for developing geothermal power generation capacity in Rwanda. It outlines a comprehensive strategy to leverage Rwanda's significant geothermal resources to meet growing electricity demand, reduce dependence on fossil fuels dependence, and support national development goals.

7.1 Key Documents

- **BUSINESS_PLAN.md** - Comprehensive business plan for geothermal power generation in Rwanda
- **SUMMARY.md** - Research findings on geothermal energy potential, prospect zones, exploration history, and development plans
- **HOT_SPRINGS_ENERGY_PROFITABILITY.md** - Profitability and ROI analysis of harnessing Rwanda's hot springs energy
- **PROJECT_TIMELINE.md** - Consolidated implementation timeline and specific activities
- **RISK_REGISTER.md** - Detailed risk assessment and mitigation strategies
- **JICA.md** - Findings from the Japan International Cooperation Agency project on Rwanda's Electricity Development Plan
- **MININFRA.md** - Summary of Rwanda's Ministry of Infrastructure energy sector information

7.2 Overview

The Shyuha project focuses on developing geothermal power generation through a phased approach that begins with resource validation and pilot projects, progressing to commercial-scale power plants. The business plan presents a compelling investment case based on Rwanda's significant geothermal potential (conservative estimates of up to 100 MW, with higher estimates exceeding 700 MW) and the opportunity to provide clean, reliable baseload power to support the nation's energy access and climate goals.

7.3 Business Model

The venture will operate as an Independent Power Producer (IPP) developing, owning, and operating geothermal power plants under long-term Power Purchase Agreements (PPAs) with Rwanda Energy Group (REG) or direct sales to industrial customers. Revenue will be primarily generated through electricity sales at regulated tariffs.

7.4 Implementation Approach

The plan follows a three-phase approach: 1. **Phase 1 (Foundation Building)**: Resource validation, partnerships, and direct-use pilots 2. **Phase 2 (Pilot Implementation)**: Exploratory drilling and system demonstrations 3. **Phase 3 (Scale and Replicate)**: Commercial geothermal power plant deployment

7.5 Opportunity

Geothermal power offers significant advantages for Rwanda: - Baseload capacity with 90-95% availability (complements variable hydro and solar) - Low operating costs after capital investment - Long lifespan (25-30+ years) - Reduced dependence on expensive diesel generation - Alignment with national goals: Vision 2020, NST1, and 38% emissions reduction pledge by 2030

Financial analysis shows attractive returns with simple payback periods of approximately 2.7 years and 25-year NPVs ranging from $55-150 million for 5-10 MW plants (after exploration risk mitigation).

Last Updated: September 2026

8 Risk Register

8.1 Scope and Method

This register covers the risks to the venture described in `BUSINESS_PLAN.md`. It replaces the narrative risk sections of the earlier `PROJECT_TIMELINE.md`, which described mitigations without assigning likelihood, impact, ownership or residual exposure.

Likelihood and impact ratings are the authors' preliminary judgement, not researched findings. They should be revisited at each gate and after Phase 1 results. Where a rating depends on a figure that does not yet exist, that is stated.

8.1.1 Rating scales

Score	Likelihood	Impact
5	Near certain (>80%)	Threatens the venture's existence
4	Likely (60–80%)	Major — delays or re-scopes a phase
3	Possible (40–60%)	Moderate — manageable within a phase
2	Unlikely (20–40%)	Minor — absorbed in normal operations
1	Remote (<20%)	Negligible

Risk score = likelihood × impact. **Bands:** 20–25 Critical · 12–16 High · 6–10 Medium · 1–5 Low.

Residual score is the exposure remaining *after* the stated mitigation, assuming it is executed.

8.2 Register

ID	Risk	Category	L	I	Score	Mitigation	Residual
R1	No commercially viable reservoir is found at any prospect — the resource is unmeasured, and the one drilled test (Karisimbi, 2006) failed	Resource	5	5	**25 Critical**	Phase 1 is structured as a falsification test with pre-registered numeric kill criteria; Phase 3 capital is contingent on the gate. Do not commit generation capital before a measured flow and temperature exist.	20 Critical

ID	Risk	Category	L	I	Score	Mitigation	Residual
R2	The venture duplicates work the state developer (EDCL) is already funded to do, and is read as such by MININFRA	Strategic	4	5	**20 Critical**	Position explicitly as the commercialisation layer, not an explorer: take a proven resource under a resource-access agreement and sell heat and power. Confirm the relationship in writing before the approach to government. See `GOVERNANCE_AND_OWNERSHIP.md`.	8 Medium

ID	Risk	Category	L	I	Score	Mitigation	Residual
R3	The venture fails GRMF eligibility — the experience criterion carries 20 of 100 points against a 70-point threshold, and a newly incorporated entity has no track record	Funding	4	4	**16 High**	Recruit a technical partner with East African geothermal delivery record; alternatively apply as a partner to EDCL's programme rather than as a standalone applicant. Confirm current GRMF terms before submission — the terms cited here are from the developer manual and may have been revised.	8 Medium

ID	Risk	Category	L	I	Score	Mitigation	Residual
R4	No customer can be secured for direct-use heat, and the avoided-cost case does not hold because the target customer does not currently burn expensive fuel	Market	4	4	**16 High**	Audit at least three named prospective customers' current fuel type and spend before designing the pilot. Reframe Phase 2 as a grant-funded demonstration if no customer signs a term sheet.	12 High

ID	Risk	Category	L	I	Score	Mitigation	Residual
R5	The generation tariff is unknown, so project revenue cannot be modelled	Revenue	4	4	**16 High**	Treat tariff as a model *output* (a breakeven) rather than an input until a RURA REFIT level or tender result is confirmed. See `../FINANCIAL_MODEL.md`.	8 Medium

ID	Risk	Category	L	I	Score	Mitigation	Residual
R6	Drilling costs exceed budget — no Rwandan drilling cost data exists, and costs are benchmarked from Kenya	Cost	4	3	**12 High**	Benchmark from Kenyan actuals with an explicit contingency; structure drilling as a fixed-price contract where possible; seek partial risk guarantee cover for exploration wells.	6 Medium

ID	Risk	Category	L	I	Score	Mitigation	Residual
R7	Currency movement — costs are largely in RWF, and any power revenue is likely USD-denominated	Financial	3	4	**12 High**	Establish PPA currency at term-sheet stage; seek local-currency financing for RWF-denominated cost base; state the FX assumption explicitly rather than leaving it implied.	6 Medium

ID	Risk	Category	L	I	Score	Mitigation	Residual
R8	Land access cannot be secured, or the prospects are constrained by proximity to Volcanoes National Park	Land and environment	3	4	**12 High**	Confirm tenure and protected-area status for each prospect before committing appraisal spend. Engage REMA early. Northern prospects sit in or near the national park.	6 Medium

ID	Risk	Category	L	I	Score	Mitigation	Residual
R9	Permitting and environmental approval take longer than planned	Regulatory	4	3	**12 High**	Early engagement with REMA and local authorities; sequence licence application ahead of the construction decision, not alongside it.	6 Medium

ID	Risk	Category	L	I	Score	Mitigation	Residual
R10	Reservoir declines faster than expected, requiring more make-up wells than budgeted	Technical	3	3	9 Medium	Include a make-up well reserve in the financial model from the outset — its omission was a principal defect of the earlier analysis. Plan reservoir monitoring from the first well.	6 Medium

ID	Risk	Category	L	I	Score	Mitigation	Residual
R11	The plan itself rests on unverified figures that turn out to be wrong	Data integrity	4	3	**12 High**	Every figure is tagged Measured, Benchmarked or `[ASSUMED - VERIFY]` in `DATA_GAPS_AND_ASSUMPTIONS.md`. No figure is presented as researched when it is not.	4 Low

ID	Risk	Category	L	I	Score	Mitigation	Residual
R12	Community opposition or failure to secure social licence at the prospect	Social	2	4	8 Medium	Early and sustained community engagement; local employment commitments; transparent benefit-sharing. Direct-use pilots create visible local benefits earlier than drilling does.	4 Low

ID	Risk	Category	L	I	Score	Mitigation	Residual
R13	A technical partner or contractor fails to perform	Execution	3	3	9 Medium	Competitive procurement; milestone-linked payment; retain an independent engineer for drilling supervision.	6 Medium
R14	Political or sovereign change disrupts the programme	Political	2	5	10 Medium	Align the venture with stated national policy so it survives administrative change; avoid single-counterparty dependence.	8 Medium

ID	Risk	Category	L	I	Score	Mitigation	Residual
R15	Carbon credit revenue is assumed but does not materialise	Revenue	4	2	8 Medium	Exclude carbon revenue from the base case entirely; treat it as upside only, and only if a registry and methodology are confirmed.	2 Low

8.3 The Three Risks That Should Decide the Approach

R1 — the resource. Everything else is secondary. Rwanda has drilled exactly one documented test (Karisimbi, 2006) and it failed. If Phase 1 also fails, the venture's scope narrows to direct-use heat only, and the plan says so up front rather than burying it.

R2 — duplication with EDCL. This is the risk most likely to end the venture before it starts, and it is entirely avoidable. EDCL already holds GRMF funding for surface studies including direct-use feasibility at Gisenyi, and separately for exploration drilling at Kinigi. Approaching MININFRA with a proposal to explore those same prospects invites a one-line dismissal. Approaching as the party that commercialises what EDCL proves is a different and additive proposition.

R4 — the direct-use customer. Phase 2's entire revenue case assumes a customer who burns expensive fuel today and will sign a long-term heat agreement. Neither is established. The earlier repository analysis quoted a 0.5–1.5 year payback for a 500 kW thermal greenhouse system; that figure is arithmetic, not a forecast, and it is the same class of error as the electrical payback it replaced.

8.4 Review Cycle

Trigger	Action
Phase 1 gate reached	Re-rate R1 and R6 on measured results; escalate or close Phase 3
Quarterly	Review R2–R5 and R7 with the board
Annually	Full re-rate, aligned with Rwanda's national planning cycle
Ad hoc	On material policy, financing or technical change

8.5 Sources

1. GRMF Developer Manual (PDF)
2. ThinkGeoEnergy - Rwanda seeks drilling grant from GRMF

Note: Likelihood and impact ratings are the authors' preliminary judgement and are not sourced findings. GRMF terms cited are drawn from the developer manual and must be re-confirmed against the current application round before any submission.

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9 Solar Power Generation ROI Analysis: 100kW to 10MW Systems for Kigali City

9.1 Executive Summary

This analysis examines the return on investment (ROI) for solar photovoltaic (PV) power generation systems ranging from 100kW to 10MW capacity, specifically targeting electricity supply to Kigali City, Rwanda. Based on available data from Rwanda-specific solar calculators and utility-scale project references, we can estimate economic viability for different system sizes.

9.2 Key Assumptions (Based on Rwanda-Specific Data)

From the AfroTools Rwanda Solar Panel ROI Calculator:

- **Electricity tariff**: 182 RWF/kWh (approximately $0.16 USD/kWh at 2026 exchange rates)
- **Fuel cost comparison**: 1,700 RWF/L (diesel for comparison)
- **Installation cost**: 750,000 RWF/kW (approximately $650 USD/kW)
- **Solar yield**: 4.8 peak sun hours/day (excellent solar resource)
- **System lifetime**: 25 years
- **Degradation rate**: 0.5% per year
- **Operations & maintenance**: 1% of CAPEX annually

9.3 Economic Analysis by System Size

9.3.1 100kW System

- **CAPEX**: 100 kW × 750,000 RWF/kW = 75,000,000 RWF (~$65,000 USD)
- **Annual generation**: 100 kW × 4.8 h/day × 365 days × 0.85 (system losses) 148,920 kWh/year
- **Annual revenue**: 148,920 kWh × 182 RWF/kWh 27,103,440 RWF (~$23,500 USD)
- **Simple payback**: ~2.8 years (based on scaling from AfroTools 5kW data showing 2.6-3.8 years)
- **25-year savings**: Approximately 425-635 million RWF (~$370,000-$550,000 USD)

9.3.2 500kW System

- **CAPEX**: 500 kW × 750,000 RWF/kW = 375,000,000 RWF (~$325,000 USD)
- **Annual generation**: ~744,600 kWh/year
- **Annual revenue**: ~135,517,200 RWF (~$117,500 USD)
- **Simple payback**: ~2.8 years
- **25-year savings**: Approximately 2.1-3.2 billion RWF (~$1.8-$2.8 million USD)

9.3.3 1MW System

- **CAPEX**: 1 MW × 750,000 RWF/kW = 750,000,000 RWF (~$650,000 USD)
- **Annual generation**: ~1,489,200 kWh/year
- **Annual revenue**: ~271,034,400 RWF (~$235,000 USD)
- **Simple payback**: ~2.8 years
- **25-year savings**: Approximately 4.2-6.3 billion RWF (~$3.6-$5.5 million USD)

9.3.4 5MW System

- **CAPEX**: 5 MW × 750,000 RWF/kW = 3,750,000,000 RWF (~$3.25 million USD)
- **Annual generation**: ~7,446,000 kWh/year
- **Annual revenue**: ~1,355,172,000 RWF (~$1.175 million USD)
- **Simple payback**: ~2.8 years
- **25-year savings**: Approximately 21-31.5 billion RWF (~$18-$27.5 million USD)

9.3.5 10MW System (Reference: Gesto Energy Project)

- **Reported CAPEX**: USD 37.6 million for 10MW (higher than our estimate due to:
  - Land acquisition and preparation
  - Grid interconnection costs
  - Higher quality components
  - Project development and financing costs)
- **Annual generation**: ~41,900,000 kWh/year (10MW × 4.8 × 365 × 0.85)
- **Annual revenue**: ~7,625,800,000 RWF (~$6.6 million USD)
- **Simple payback**: ~5.7 years (based on reported $37.6M CAPEX)
- **25-year savings**: Approximately 114-171 billion RWF (~$99-$149 million USD)

9.4 Comparison with Conventional Power Generation

9.4.1 Diesel Generation Comparison

- **Diesel generator efficiency**: ~3-4 kWh/L
- **Fuel cost**: 1,700 RWF/L
- **Effective cost**: ~425-567 RWF/kWh
- **Solar advantage**: Solar at 182 RWF/kWh provides 65-75% cost savings vs diesel

9.4.2 Grid Electricity Comparison

- **Current grid mix**: ~47% hydro, remainder thermal (diesel/heavy fuel oil)
- **Grid electricity cost**: Similar to tariff rate (182 RWF/kWh) for commercial/industrial
- **Solar advantage**: Locks in electricity cost for 25 years vs potential tariff increases

9.5 Sensitivity Analysis

9.5.1 Key Variables Affecting ROI:

1. **Installation cost**: ±20% changes payback by ±0.6 years
2. **Solar yield**: ±0.5 h/day changes payback by 0.3 years
3. **Electricity tariff**: ±20% changes payback by 0.6 years
4. **O&M costs**: ±50% changes payback by ±0.2 years

9.6 Recommendations for Kigali City Implementation

9.6.1 Phased Approach:

1. **Pilot Phase (100kW-500kW)**:
   - Target: Government buildings, schools, hospitals

- Benefit: Demonstrates technology, builds local expertise
- Expected payback: 2.6-3.8 years

2. **Commercial Phase (1MW-5MW)**:
   - Target: Industrial zones, commercial complexes
   - Benefit: Significant cost savings for large consumers
   - Expected payback: 2.6-3.8 years
3. **Utility Phase (5MW-10MW+)**:
   - Target: Grid-connected utility scale
   - Benefit: Peak shaving, renewable energy targets
   - Expected payback: 4-6 years (including grid interconnection)

9.6.2 Financing Considerations:

- **Available incentives**: Potential for green climate funds, World Bank support
- **PPA models**: Power Purchase Agreements can eliminate upfront CAPEX
- **Green bonds**: Increasing availability for renewable energy projects
- **Development finance**: AfDB, IFC, FMO offer favorable terms for African solar

9.7 Conclusion

Solar PV power generation in the 100kW to 10MW range presents attractive ROI for Kigali City applications, with simple payback periods ranging from 2.6 to 5.7 years depending on system size and specific project costs. The excellent solar resource (4.8 kWh/m²/day), high electricity tariffs, and declining solar costs make solar PV increasingly competitive with conventional generation.

For immediate implementation, 100kW-500kW systems targeting high-tariff commercial and institutional customers offer the fastest returns. Larger utility-scale projects (5MW+) require more sophisticated financing but provide substantial long-term savings and contribute significantly to Rwanda's renewable energy goals.

9.8 Sources

1. Rwanda Solar Panel ROI Calculator | AfroTools
2. Gesto Energy | Rwanda 10MW Solar PV Development
3. Utility-Scale Solar Cost Per MW 2026: EPC Bid Guide
4. Solar - Rwanda Energy Group
5. AFROTools Estimate solar payback in Rwanda

Note: All currency conversions based on approximate 2026 exchange rates. Actual project costs may vary based on specific site conditions, financing structures, and market fluctuations.

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10 Energy Required to Pump Water to Kigali City

10.1 Overview

This document analyzes the energy requirements for pumping water to Kigali City, Rwanda, focusing on the elevation difference between water sources and the city, and providing calculations for the electrical energy needed per cubic meter of water delivered.

10.2 Key Elevation Data

- **Kigali City elevation**: 1,476 meters above sea level [Source: Elevation.maplogs.com]
- **Primary water source (Lake Mugesera/Karenge)**: 1,300 meters above sea level [Source: Wikipedia - Lake Mugesera]
- **Static elevation head**: 1,476 m - 1,300 m = **176 meters**

10.3 Fundamental Physics of Water Pumping

The theoretical energy required to lift water is calculated using the formula:

Energy (joules) = Volume (m^3) × Density (kg/m^3) × Gravity (m/s^2) × Head (m)

Where: - Density of water = 1,000 kg/m^3 - Gravity = 9.81 m/s^2

To convert joules to kilowatt-hours (kWh): **Energy (kWh) = Energy (joules) / 3,600,000**

Combining these, the theoretical energy per cubic meter per meter of head is: **Energy ($kWh/m^3/m$) = (1,000 × 9.81) / 3,600,000 = 0.002725 $kWh/m^3/m$**

10.4 Practical Energy Calculation (Including System Efficiency)

Real-world pumping systems have energy losses due to: - Pump inefficiency (typically 60-85% efficient) - Motor inefficiency (typically 85-95% efficient) - Pipe friction losses - Fitting and valve losses

The practical formula is: **Energy (kWh/m^3) = (Head in meters) / (367 × Overall Efficiency)**

Where 367 is derived from: 3,600,000 / (1,000 × 9.81)   367

Typical overall efficiency for water pumping systems (pump + motor + controls): **60-75%**

10.5 Energy Calculation for Kigali's Static Head

Using Kigali's 176-meter static head:

10.5.1 At 60% Overall Efficiency:

Energy = 176 / (367 × 0.60) = 176 / 220.2 = **0.80 kWh/m³**

10.5.2 At 70% Overall Efficiency:

Energy = 176 / (367 × 0.70) = 176 / 256.9 = **0.68 kWh/m³**

10.5.3 At 75% Overall Efficiency:

Energy = 176 / (367 × 0.75) = 176 / 275.25 = **0.64 kWh/m³**

10.6 Additional Energy Components: Friction Losses

The static head calculation above only accounts for lifting water against gravity. Actual systems require additional energy to overcome:

1. **Pipe friction losses**: Depends on pipe length, diameter, material, and flow rate
2. **Fitting losses**: Elbows, valves, tees, etc.
3. **Entry/exit losses**: At intake and discharge points

For municipal water supply systems, friction losses typically add **20-50%** to the static head requirement, depending on: - Distance from source to city - Pipe diameter and condition - Flow rate (higher flow = higher friction losses) - System age and maintenance

10.6.1 Estimated Total Head for Kigali System

Assuming moderate friction losses (30% additional head): - Static head: 176 meters - Friction losses: 176 × 0.30 = 53 meters - **Total dynamic head**: ~229 meters

Revised energy calculation (at 70% efficiency): Energy = 229 / (367 × 0.70) = 229 / 256.9 = **0.89 kWh/m³**

10.7 Context: Water Consumption and Energy Implications

10.7.1 Kigali's Water Demand

- Estimated daily water consumption: ~100,000 m³/day (varies by source and growth)
- Annual consumption: ~36.5 million m³/year

10.7.2 Annual Electricity Requirement for Pumping

At 0.89 kWh/m³: - Daily: 100,000 m³ × 0.89 kWh/m³ = **89,000 kWh/day** - Annual: 36,500,000 m³ × 0.89 kWh/m³ = **32,485,000 kWh/year** (~32.5 GWh/year)

10.7.3 Cost Implications

Using Rwanda's electricity tariff (~182 RWF/kWh or ~$0.16 USD/kWh): - Daily pumping cost: 89,000 kWh × 182 RWF/kWh = **16,198,000 RWF/day** (~$14,240 USD/day) - Annual pumping cost: 32,485,000 kWh × 182 RWF/kWh = **5,912,270,000 RWF/year** (~$5.2 million USD/year)

10.8 Comparative Analysis

10.8.1 Comparison to Other Cities

- **Johannesburg, South Africa**: ~0.4-0.6 kWh/m³ (lower elevation gain)
- **Denver, Colorado, USA**: ~0.3-0.5 kWh/m³ (Rocky Mountain foothills)
- **Mexico City**: ~0.8-1.2 kWh/m³ (significant elevation gain ~1,000m)
- **La Paz, Bolivia**: ~1.2-1.8 kWh/m³ (very high elevation ~3,600m)

Kigali's requirement of ~0.68-0.89 kWh/m³ reflects its moderate elevation gain of ~176m from Lake Mugesera.

10.8.2 Comparison to Alternative Water Sources

If Kigali were to use sources at different elevations: - **From Nyabarongo River (~1,350m)**: Head = 126m → ~0.49-0.64 kWh/m³ - **From deeper wells (~1,400m)**: Head = 76m → ~0.30-0.39 kWh/m³ - **From higher sources (~1,500m)**: Would require less pumping or could use gravity flow

10.9 Optimization Opportunities

10.9.1 1. System Efficiency Improvements

- **High-efficiency pumps**: Modern pumps can reach 85-90% efficiency
- **Premium efficiency motors**: IE3/IE4 motors 92-96% efficient
- **Variable frequency drives (VFDs)**: Match pump speed to demand, saving 20-50% energy
- **Regular maintenance**: Clean impellers, aligned shafts, proper lubrication

10.9.2 2. Infrastructure Optimization

- **Pipe sizing**: Larger diameter reduces friction losses (higher CAPEX, lower OPEX)
- **Loop systems**: Reduce peak flows and friction losses
- **Leak detection and repair**: Reduce total volume needing pumping

- **Pressure management**: Reduce excess pressure in distribution network

10.9.3 3. Renewable Energy Integration

- **Solar PV pumping**: Excellent match for daytime water demand
- **Hydropower recovery**: Energy recovery from pressure reduction valves
- **Grid-tied systems with net metering**: Offset pumping costs with renewable generation

10.10 Comparison with Solar PV Generation Potential

From the SOLAR_ROI_ANALYSIS in this repository: - **100kW solar system** in Kigali generates ~148,920 kWh/year - This could pump approximately: 148,920 kWh/year ÷ 0.89 kWh/m³ = **167,000 m³/year** - Or about **457 m³/day** - roughly 0.46% of Kigali's estimated daily demand

10.10.1 Solar-Powered Pumping Scenarios

- **1MW solar system**: Could pump ~4,570 m³/day (~4.6% of demand)
- **5MW solar system**: Could pump ~22,850 m³/day (~23% of demand)
- **10MW solar system**: Could pump ~45,700 m³/day (~46% of demand)

This demonstrates that solar PV could offset a significant portion of Kigali's water pumping energy requirements, particularly during daylight hours when both solar production and water demand peak.

10.11 Recommendations for Kigali Water Supply

1. **Conduct detailed hydraulic modeling** to determine exact friction losses
2. **Implement energy audits** of existing pumping stations
3. **Prioritize high-efficiency equipment** in upgrades and replacements
4. **Consider solar PV integration** for daytime pumping operations
5. **Implement VFDs** on all major pumping stations
6. **Explore gravity-fed options** from higher elevation sources where feasible
7. **Monitor and benchmark** energy intensity (kWh/m³) regularly for continuous improvement

10.12 Sources

1. Elevation of Kigali, Rwanda - MAPLOGS: https://elevation.maplogs.com/poi/kigali_rwanda.37590.html
2. Lake Mugesera elevation - Wikipedia: https://en.wikipedia.org/wiki/Lake_Mugesera
3. Karenge Drinking Water Supply System - Wikipedia: https://en.wikipedia.org/wiki/Karenge_Drinking_
4. Engineering ToolBox - Water Pumping Costs: https://www.engineeringtoolbox.com/water-pumping-costs-d_1527.html

5. Standard pumping energy calculation formulas (hydraulic engineering principles)

Note: Actual energy consumption will vary based on specific system design, maintenance condition, flow rates, and local factors. The calculations above provide a reasonable engineering estimate for planning and analysis purposes.
